

ERP Research Methodology and Next Steps

1. ERP Research Methodology

The team agreed that the new ERP should be designed with a zero-bias approach toward the existing ERP system. The existing system should not be used as the starting point for identifying requirements because doing so could result in developing a replica of the current system with only minor changes. The objective is to develop a fresh, market-benchmarked ERP product that is based on current business requirements, industry practices, and market research rather than on the limitations or structure of the existing system.

The proposed product strategy is to follow an 80% core product, 10% configuration, and 10% customization approach. Approximately 80% of the ERP should consist of standard core functionality that can be reused across clients. Around 10% should be handled through configuration so that client-specific requirements can be accommodated without changing the core product. The remaining 10% can be addressed through customization when a requirement cannot be handled through standard functionality or configuration. This approach is intended to avoid rigid client-specific development, which can make the product difficult to maintain and prevent it from being easily deployed for new clients.

The research and product-definition process has been finalized as Market Research, Analysis, Business Requirement Document (BRD), Functional Requirement Document (FRD), Prototype, and Fundamental Test Cases. Market research will be conducted through multiple iterations and can be supported by AI tools. The information collected during market research will then be analyzed to identify common capabilities, industry requirements, differentiating features, and potential gaps.

Once the analysis is completed, the findings will be converted into a foundational BRD. The BRD will define the business requirements and the reasons behind them. After stakeholder approval of the BRD, the team will proceed with the FRD, which will define the detailed functional behavior of the system. The approved requirements will then be used to create prototypes and subsequently develop fundamental test cases.

2. Application of the 5W1H Framework

The team is using the 5W1H framework to ensure that requirements are properly understood before solutions are designed. The framework consists of What, Why, Where, Who, When, and How.

The team should first determine what the requirement or business problem is and why it is required. It should then identify where the requirement applies, who is involved in the process, and when the activity occurs or gets triggered. Only after these questions have been properly analyzed should the team determine how the requirement should be implemented.

Jumping directly to the “How” can result in the team designing a technical solution before fully understanding the business problem. Therefore, sufficient time should be spent analyzing the five “W”

questions before moving to implementation. This approach will help keep the entire team aligned and reduce the possibility of developing functionality that does not address the actual business requirement.

3. Identification of ERP Modules

Through the unbiased market research process, the team has identified 20 major ERP modules. These modules cover the major business areas expected in a comprehensive ERP product.

The identified modules are Finance and Accounting, Controlling and Cost Management, Budget Management, Purchase and Procurement, Vendor Management, Inventory and Stores, Warehouse Management, Sales and Distribution, CRM, HRMS and Payroll, Asset Management, Maintenance Management, Project Management, Manufacturing and Production Management, Quality Management, Supply Chain and Logistics, Contract Management, Customer Service and Help Desk, MIS and BI Dashboard, and Administration, Workflow, Security, and Integration.

Each major module has approximately seven to eight or more sub-modules. Some modules, such as Finance and Accounting, contain more than 20 sub-modules. Overall, the current research covers more than 160 sub-modules.

4. Module Prioritization

The next step is to apply a weighted scoring mechanism across all 20 modules and their respective sub-modules. The purpose of this scoring exercise is to determine which modules should be included in Phase 1 and which modules can be considered for Phase 2.

The scoring mechanism will help the team prioritize modules based on factors such as business importance, market demand, industry relevance, reusability across clients, strategic value, implementation complexity, and integration requirements. The exact scoring criteria and weights should be agreed upon by the team before the final scoring exercise is performed.

The team should avoid spending significant time performing a deep analysis of modules that may ultimately not be included in Phase 1. Therefore, prioritization should be completed before proceeding with detailed research of every module.

At this stage, the existing ERP should not be used as a comparison point. Comparing the new modules with the existing ERP too early could introduce bias and influence the prioritization process. The Phase 1 and Phase 2 decisions should first be based on independent market research and business requirements.

5. Detailed Analysis of Roles and Users

The “**Who**” aspect of the 5W1H framework requires further analysis. The team needs to clearly distinguish between a user and a role within the ERP.

A user represents the actual person who accesses the system, while a role represents the responsibilities and permissions associated with a particular business function. A role determines which features a user can access, whether the user has read or write permissions, which activities the user can perform, and whether the user participates in a particular approval workflow.

Therefore, the ERP design should not only identify who uses a particular feature. It should also determine the business role responsible for the activity, the authority associated with that role, the permissions required, and the point at which the role participates in the workflow.

6. Vendor Management

Vendor Management has been identified as a high-priority area for the new ERP and requires detailed research as part of the upcoming activities.

The team should study the Government e-Marketplace (GeM) portal to understand relevant vendor registration, verification, procurement, and vendor-management processes. The research should help identify capabilities that can be considered while designing the Vendor Management module.

The Vendor Management module should be studied for areas such as vendor onboarding, KYC, verification, documentation, qualification, compliance, vendor evaluation, vendor performance, and vendor master management.

The team should also evaluate whether certain capabilities should be developed internally or obtained through existing services. For example, KYC and verification APIs are already available in the market, so the team should evaluate whether integrating an existing API would be more appropriate than developing the same capability internally.

The same build-versus-buy approach should be applied across other ERP capabilities wherever reliable and reusable third-party solutions are available.

7. Plug-and-Play Architecture

The new ERP should follow a plug-and-play and integration-friendly architecture. The ERP should not assume that every client will replace all of their existing systems.

For example, a client may want to continue using SAP for Materials Management while using the new ERP for other business functions. Therefore, the new ERP should provide suitable integration capabilities that allow it to work with existing client systems.

This approach will make the product more flexible and will support clients who already have established applications that they do not want to replace. The architecture should therefore consider APIs, external services, data exchange mechanisms, authentication, and other integration requirements as part of the product design.

8. Use of Claude and AI Tools

The team is already using Claude for research and development activities and has been encouraged to make better use of available Skills and Plugins to obtain more structured outputs.

The recommended areas include Engineering, Web Artifacts Builder, Data, and Product Management. These skills can help narrow the focus of the AI tool depending on the type of task being performed. For example, a documentation-related skill can focus the output on specifications and guides, while a product-management skill can help structure product requirements and analysis.

The team was also advised to select the model according to the complexity of the task. Faster and lighter models can be used for simple research and coding activities, while more capable models such as Sonnet 5 can be used for complex or layered problems. This approach helps avoid unnecessary token usage and ensures that the available AI capabilities are used efficiently.

9. Immediate Next Steps

The first immediate activity is to refine the existing module list using the weighted scoring mechanism. The scoring should be applied across all 20 modules and more than 160 sub-modules so that the team can clearly identify the proposed Phase 1 and Phase 2 scope.

The second activity is to prepare a foundational BRD within one week. The BRD should be sufficiently complete to present to stakeholders, including Mithil and other relevant stakeholders, for review and sign-off. The team should proceed to the FRD and prototype stages only after the business requirements have been sufficiently reviewed and approved.

The third activity is to conduct detailed research on the GeM portal for the Vendor Management module. The team should also evaluate available KYC and vendor-verification APIs and identify which capabilities should be built internally and which can be obtained through existing services.

The fourth activity is to review the Help Desk module developed for GMR. The team should study the existing use cases and functionality to determine whether the module or any of its capabilities can be reused in current ongoing systems. This activity should be coordinated with Ram.

10. Overall Direction

The overall objective is to establish a disciplined ERP product-development methodology in which requirements are first understood through unbiased market research and business analysis. The team should avoid allowing the existing ERP to dictate the structure of the new product.

The process should therefore progress from market research to analysis, BRD, stakeholder approval, FRD, prototype, and fundamental test cases. The 5W1H framework should be followed throughout this process, with particular emphasis on understanding What, Why, Where, Who, and When before deciding How.

The final product should focus on reusable core functionality, configuration-based flexibility, limited customization, and strong integration capabilities. This approach will help create an ERP that can support multiple clients while reducing dependency on client-specific development.